

UML. Activity diagram



An activity diagram is a diagram that shows the flow of transitions from one activity to another.

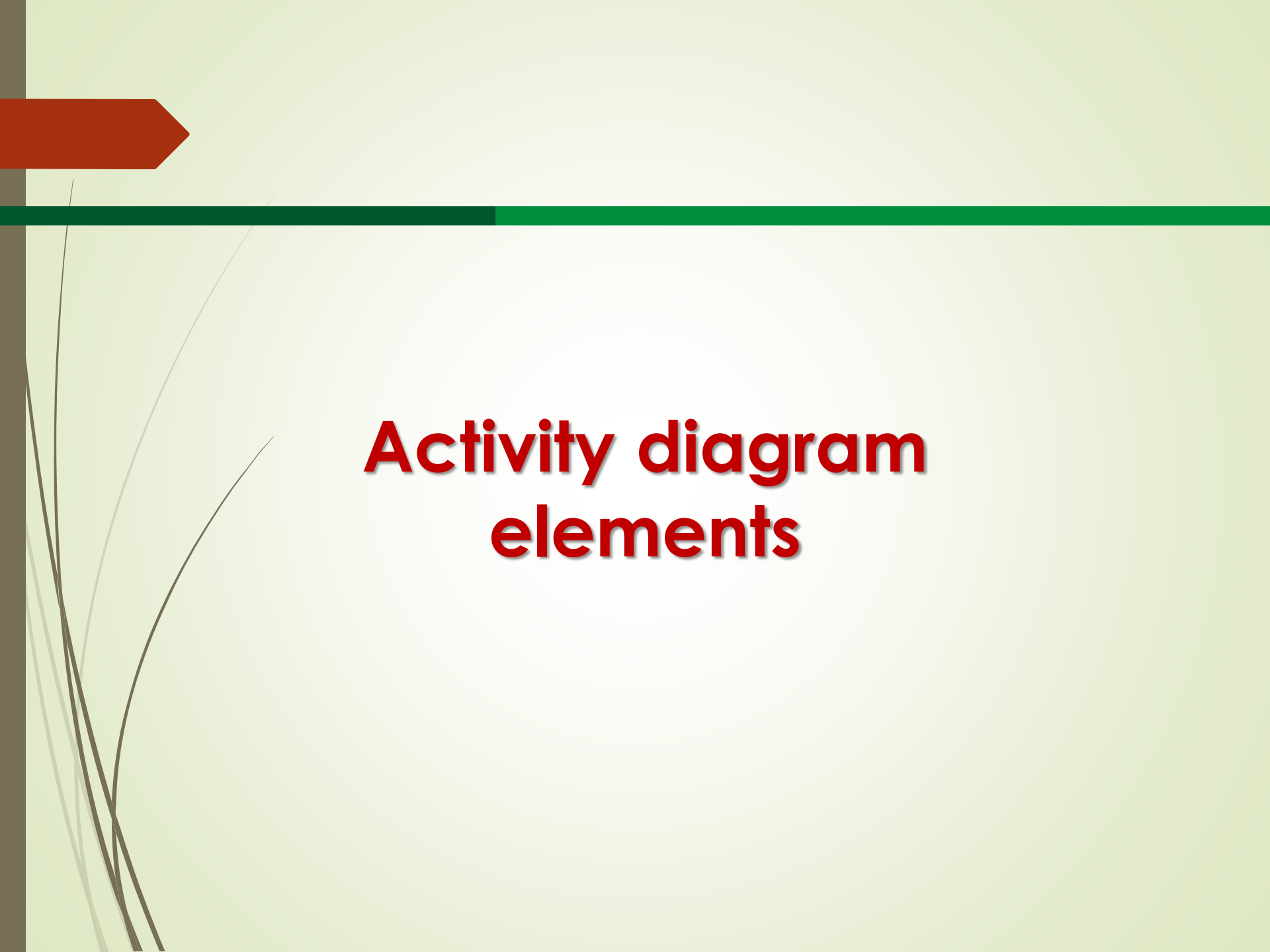
An activity diagram is designed to model the dynamic aspects of a system's behavior.

An activity diagram describes the sequence of operations performed by various system components over time and shows how the flow of control passes from one activity to another.



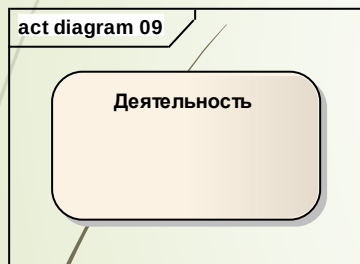
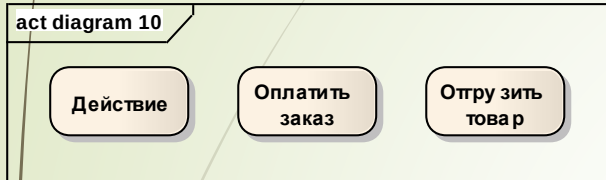
An activity diagram can contain :

- *Activity states*
- *Transitions*
- *Objects*



Activity diagram elements

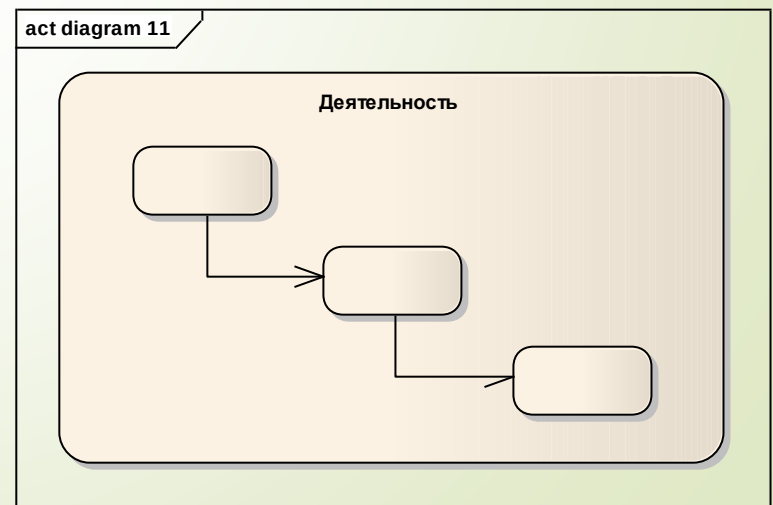
Activity status



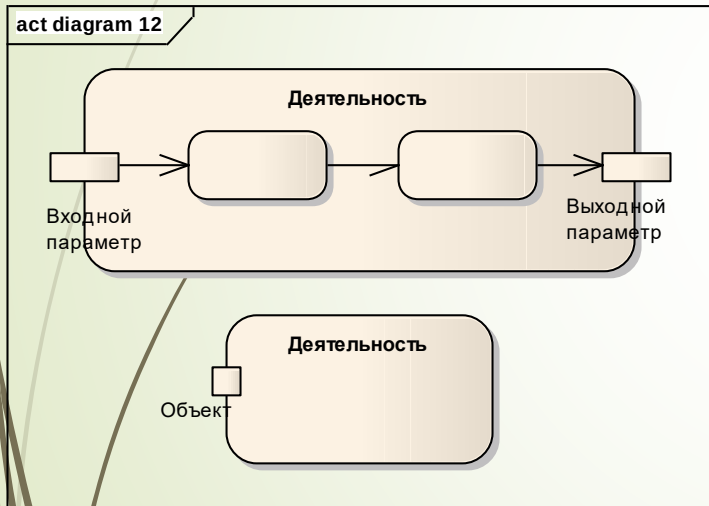
An activity may consist of a combination of smaller activities and atomic actions.

An action is an elementary unit of behavior specification that cannot be further decomposed into activities.

Activity – a non-atomic computational step within a business process that extends over time.



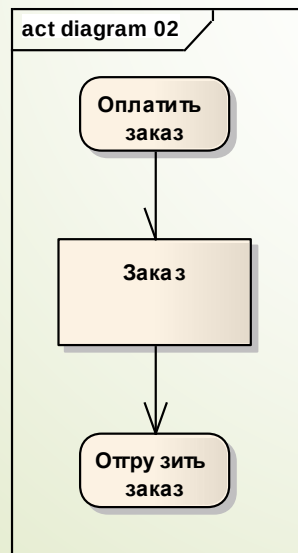
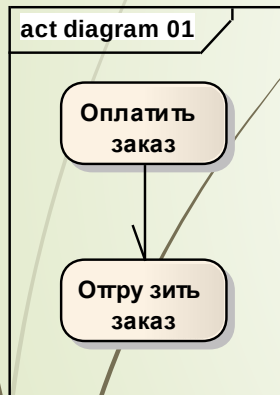
Parameters and objects of activity



An **activity parameter** serves to represent the input and output parameters of an activity.

Activity objects serve to represent data entering the activity as input or obtained as a result of its execution.

An activity edge is an abstract class for directed connections between two activity nodes.



There are two varieties of activity edges:

1. **A control flow** is an activity edge along which only control tokens are transmitted
2. **An object flow** is an activity edge along which only object or data tokens are transmitted.

Control nodes

act diagram 13

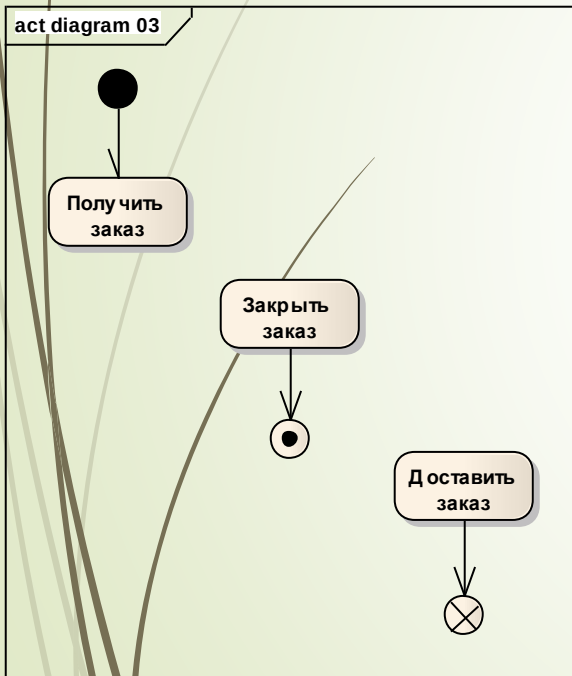


A control node is an abstract activity node designed to coordinate flows within an activity.

The following control nodes are distinguished :

- Activity initial node
- Activity final node
- Flow final node
- Decision / merge node
- Fork / join node

Control nodes



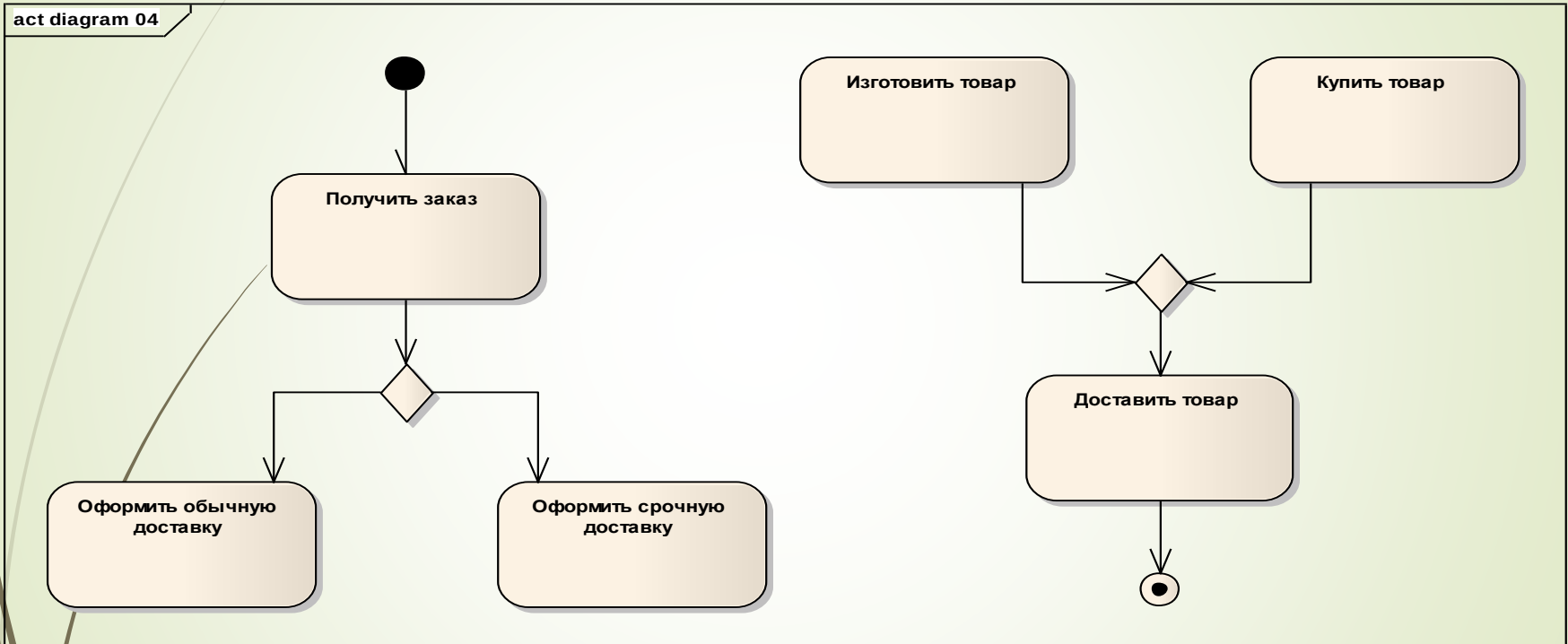
An activity initial node is a control node at which the flow begins when an activity is invoked.

An activity final node is a control node that terminates or stops all flows within an activity.

A flow final node is a final node that terminates a specific flow without terminating the activity containing it.

Control nodes

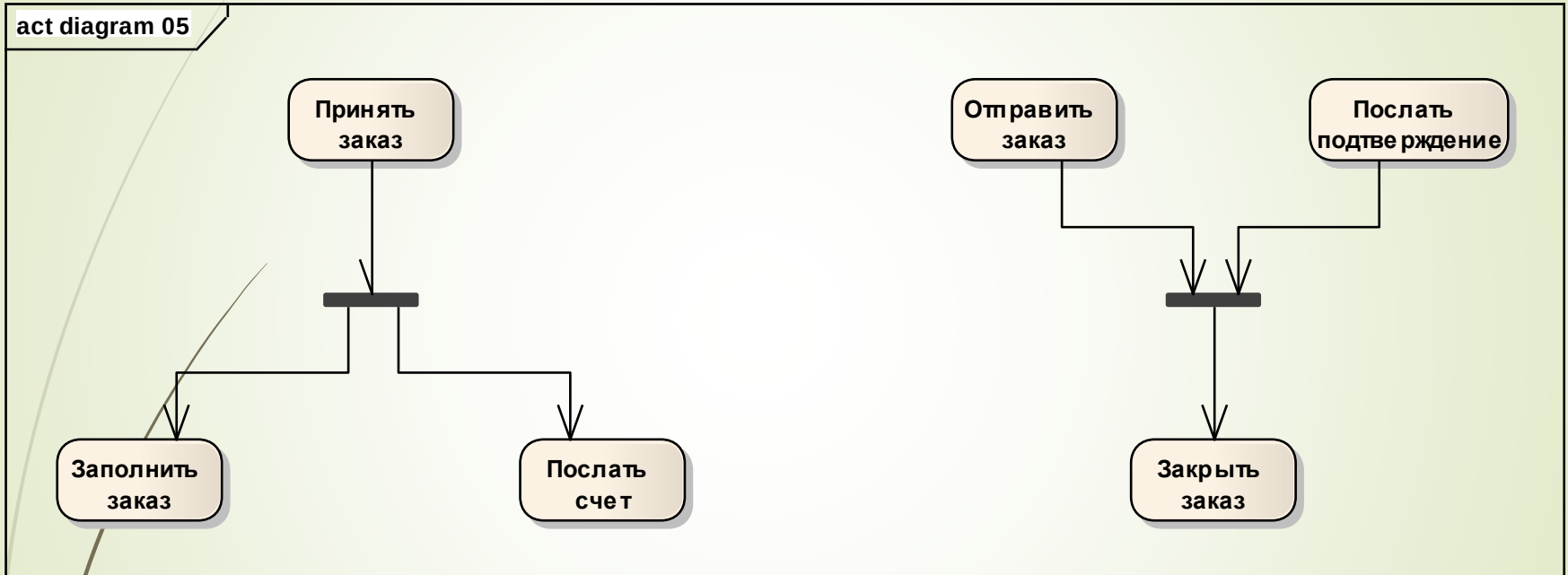
act diagram 04



A decision node is a control node that chooses between outgoing flows.

A merge node is a control node that combines multiple alternative flows.

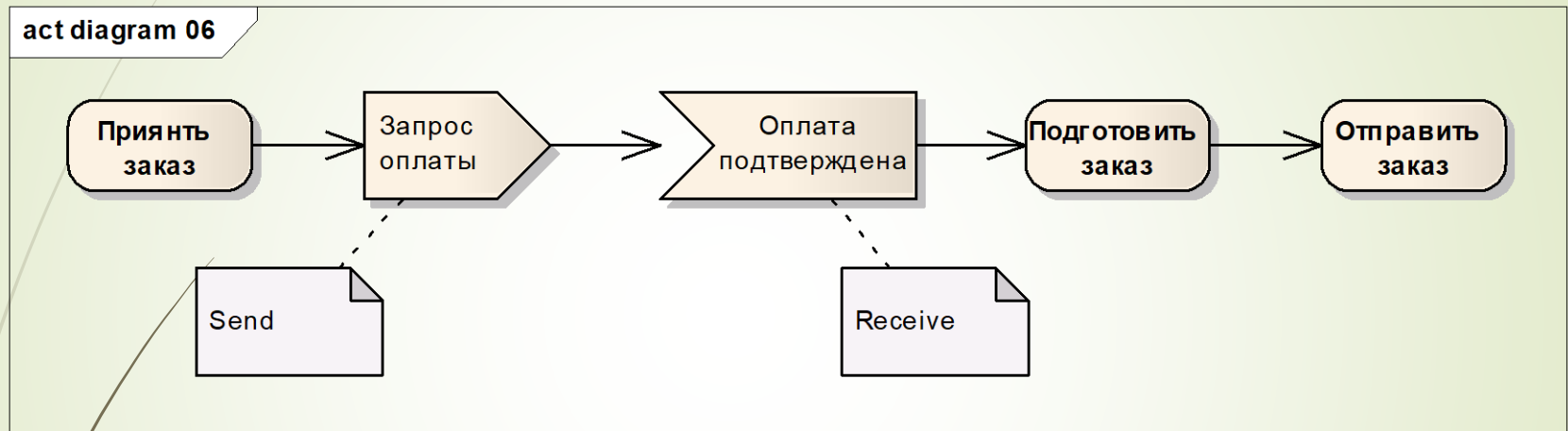
Control nodes



A fork node is a control node that splits a flow into multiple parallel flows.

A join node is a control node that synchronizes multiple flows.

Special actions

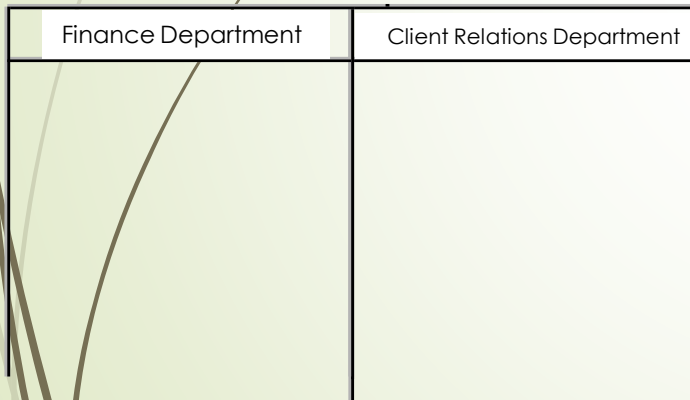


A send signal action is an action that, based on its inputs, creates a signal instance and transmits it to a target object.

A receive event action is an action that waits for the occurrence of a specific event.

Breakdown of activities

Office or
branch units

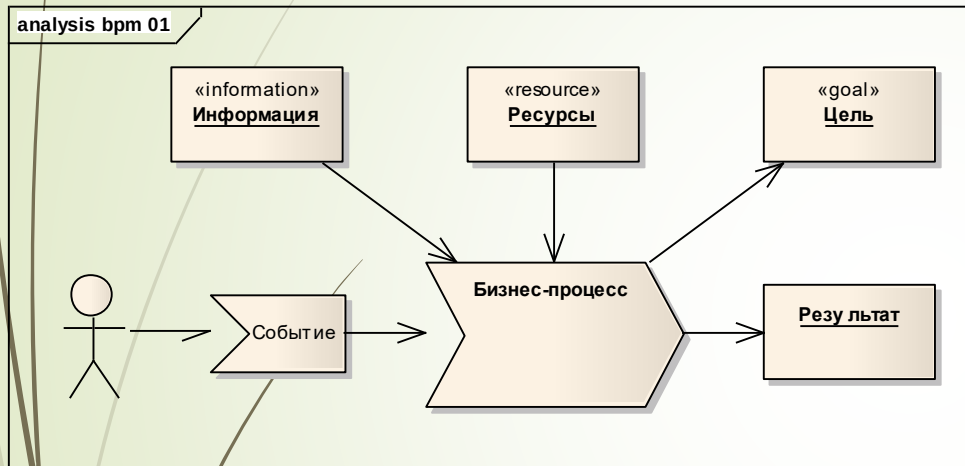


Activity partition – a model element designed to group actions that belong to the same activity and share a common characteristic.

In practice, when modeling business processes, the breakdown most frequently corresponds to organizational units.

This method of presenting activity diagrams is known as partitioning using swim lane notation.

Business analysis diagram



A business analysis diagram is a simplified activity diagram that describes high-level business processes and the general behavior of the system and its elements.

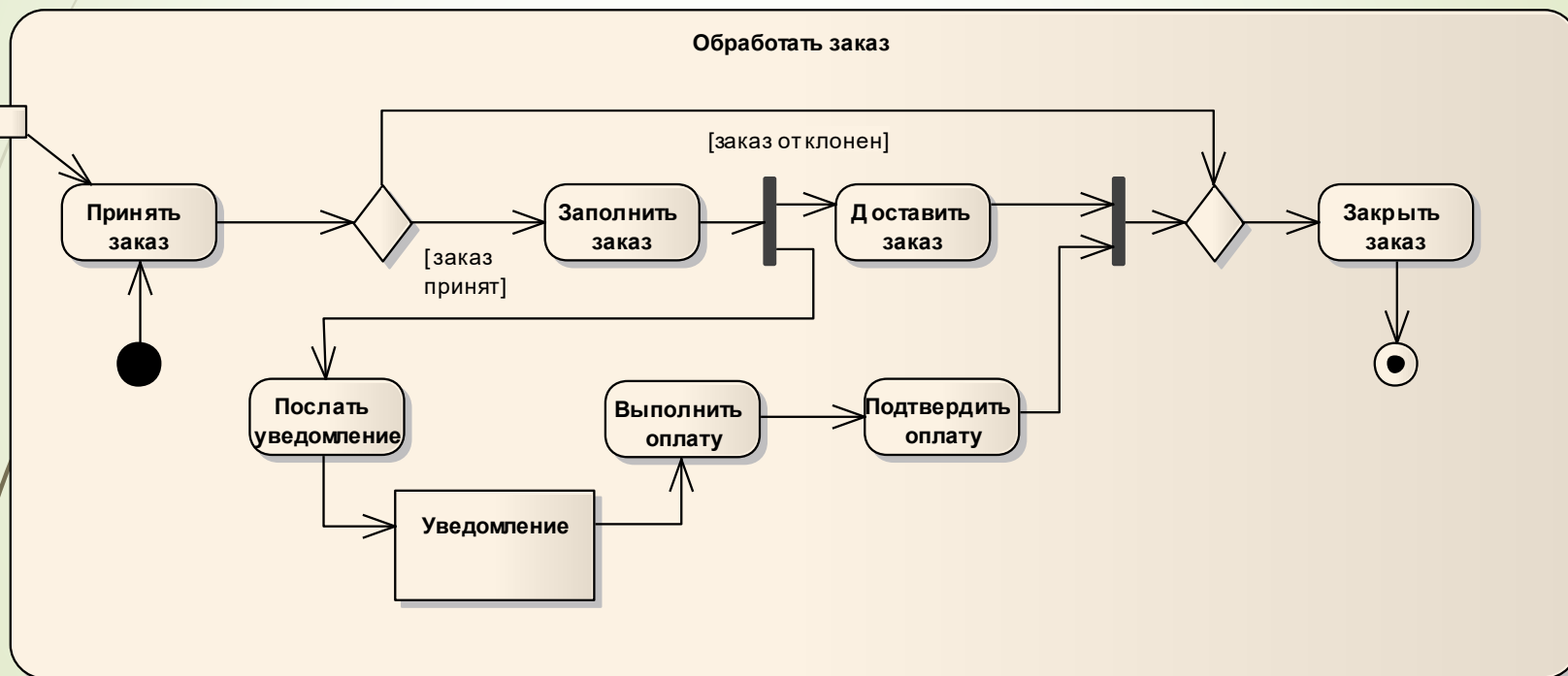
A business process model defines:

1. The process objective
2. Inputs
3. Outputs
4. Resources used
5. Actions performed in a specific order
6. The event that initiates the business process

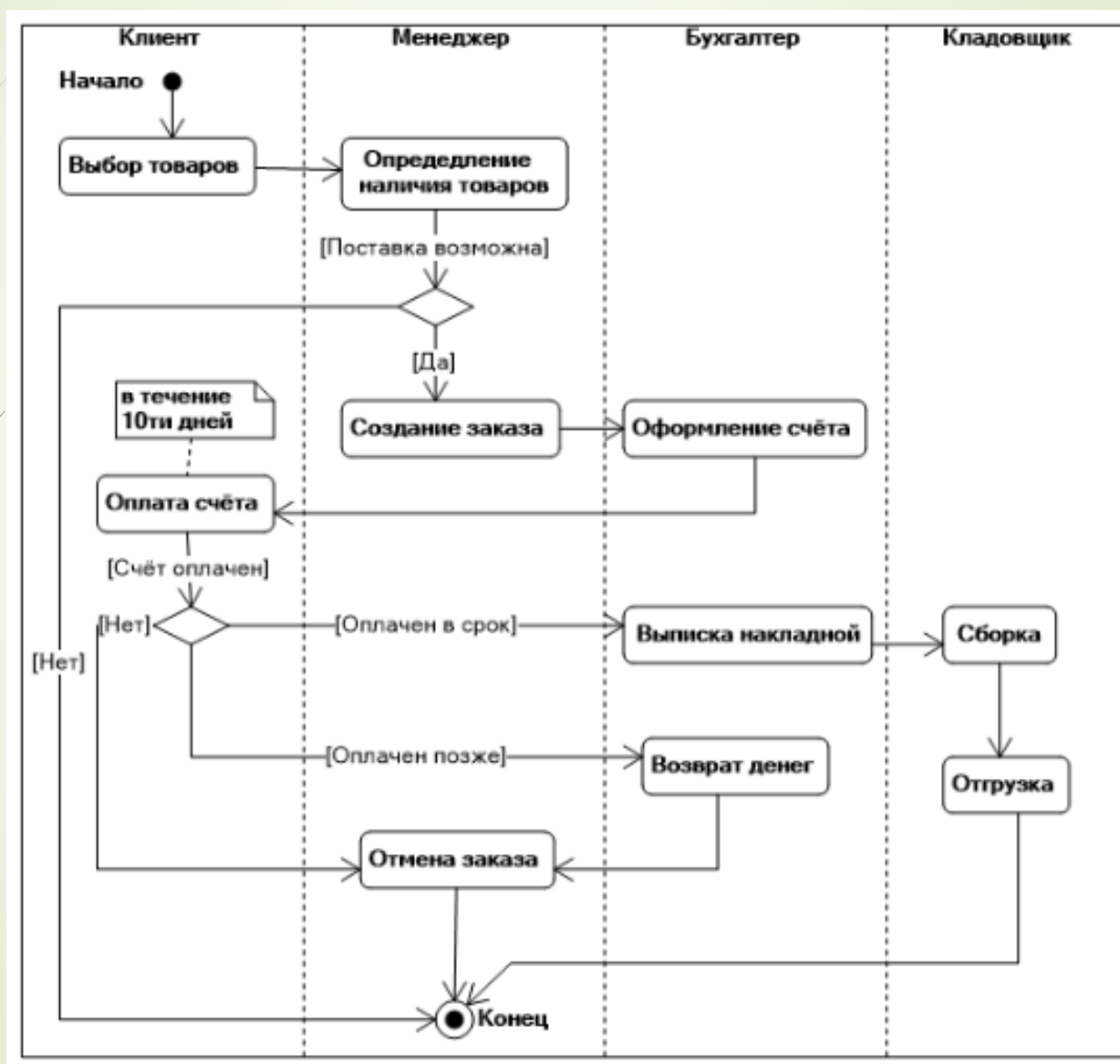


Example

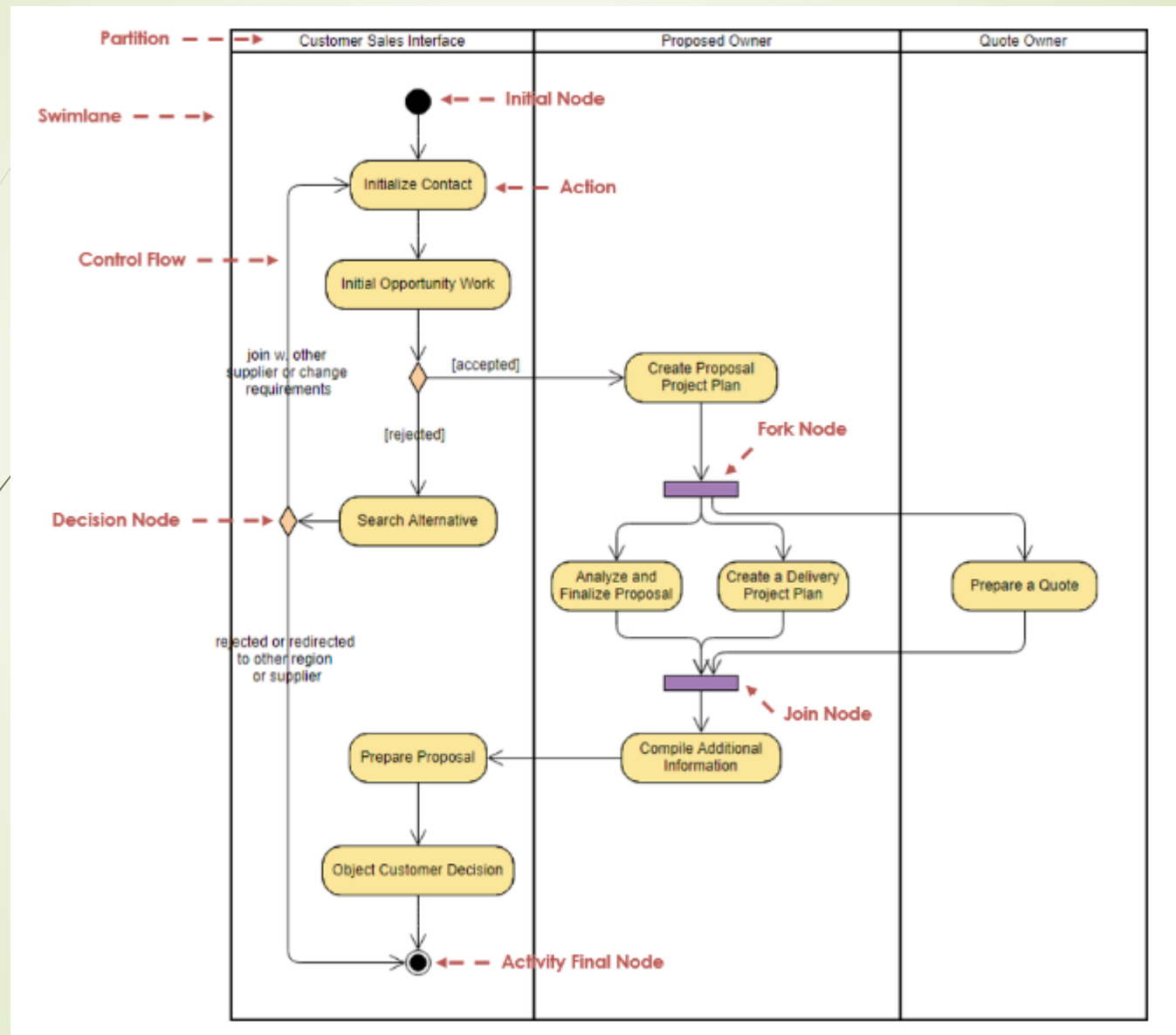
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Example



Example



1. Develop activity diagrams for your business processes
2. Expand the list of requirements identified through modeling
3. Develop a project glossary